

JOHN DOMINIQUE L. GONZALES

Software Engineer

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PROFESSIONAL SUMMARY

Programmer by craft, technologist by obsession. BS Computer Engineering graduate from Marinduque State University with proven skills in full-stack development, embedded systems, and AI-driven applications. Experienced in building web-based inventory systems for government agencies and developing computer vision solutions for real-world waste management. Seeking opportunities to apply strong analytical and problem-solving abilities in a software engineering role.

EDUCATION

Bachelor of Science in Computer Engineering

Aug 2022 - May 2026

Marinduque State University

- Best Thesis Rank 2
- Thesis on "AI-powered Smart Bin System"

WORK EXPERIENCE

Student Assistant (SPES) | Department of Agrarian Reform

June, 2025 - July, 2025

Sole Developer — PPE Management System

- Designed and built a custom frontend themed to DAR's branding and visual identity using Tailwind CSS
- Implemented a secure authentication system for role-based access control
- Developed an interactive dashboard with charts and statistics providing clear visibility into fund cluster data
- Created a normalized MySQL database schema optimized for efficient sorting, searching, and data integrity
- Built full CRUD operations for equipment data listing and management
- Implemented CSV export functionality for seamless integration with Google Sheets and external reporting

Intern | Creativoices Production

Jan 2026 – Apr 2026, 480 hours

File Management System Web App (with Yza Casimero)

- Developed backend logic for sorting and organizing desktop files programmatically
- Contributed to the frontend design and user interface of the web application

PROJECTS

AI-Powered Smart Bin with Automated Waste Segregation, Compression, and Touchless Operation, Featuring Mobile App Monitoring (Thesis)

- Trained a YOLOv8 computer vision model to classify waste into paper, plastic, plastic bottles, and other categories
- Integrated servo motors for automated lid opening and waste segregation based on classification results
- Implemented automatic air compression for paper and plastic waste to maximize bin capacity
- Developed a React Native mobile application for real-time fill-level monitoring and alerts
- Designed LED light indicators for visual status feedback

ACHIEVEMENTS

- **Best Thesis** — Rank 2, Marinduque State University
- **1st Place** — Regional Programming Competition (C++ Category)
- **2nd Place** — Regional Programming Competition (C++ Category)
- **Participant** — National Programming Competition (C++ Category)

SEMINARS & CERTIFICATES

- Blockchain is not just Crypto (Webinar)
- Formulating Competitive Marketing Strategies in the Digital Age (Webinar)
- Fundamentals of Image Processing using MATLAB and OpenCV (Seminar)
- Engineering Minds: Challenging the Sigma, Reach out, Build Bridges (Seminar)

TECHNICAL SKILLS

Languages: C++, C, Python, TypeScript, JavaScript, PHP, C# (.NET), Arduino/C++

Frontend: React.js, Next.js, React Native, Tailwind CSS

Backend & Database: Node.js, PHP, MySQL, MongoDB, Firebase

ML / AI: YOLOv8, PyTorch, TensorFlow, OpenCV

Tools & Platforms: Docker, Git, GitHub

Embedded Systems: Arduino, IoT sensors, servo motor control